

Homework 1

Due January 28th

Problems:

1. For a **non-inverting amplifier** with $R_1 = 10 \text{ k}\Omega$ and $R_2 = 400 \text{ k}\Omega$ calculate the value for a where the amplifier closed-loop gain will have an error of 10% as compared to if $a = \infty$.
2. Repeat the previous for a **inverting amplifier**.
3. Book Problem 1.27. Assume $a = \infty$
4. Book Problem 1.37. Assume $a = \infty$. Try to solve with only two op-amps; the solution is trivial when using 3 op-amps. The 3 op-amp solution will be accepted, but the problem will only receive half credit.
5. For a generic negative feedback system with an error amplifier gain of $a_\varepsilon = 50^3 \text{ V/V}$ and a closed loop gain of $A = 40$, find:
 - (i) The feedback gain, b , of the system,
 - (ii) Find A for if a_ε decreases by 5% and 50%. Note that a drop of 50% is not considered a “small change.”
 - (iii) Repeat the above assuming the initial value for a_ε was 50^2 V/V .
6. Book 1.49
7. Book 1.52